

REPORT FRONT SHEET

MOLECULAR BIOLOGY, Master Course EXPERIMENTAL CELL BIOLOGY

NAME OF EXPERIMENT: Exercise 2 A/B week 1-2: Cell Cycle Analysis and Hoechst 33342 / Alamar Blue 96 well assays

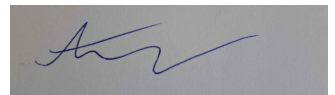
DATE FOR PERFORMANCE OF EXPERIMENT: 15-06-2026 -- 26-06-2026

TEACHER: Camilla Michelle Thorup Andersen

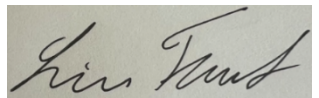
TEAM Green GROUP NUMBER: 3

Names and signatures

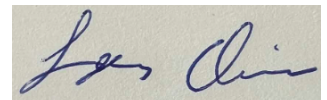
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DATE(S) FOR HANDING IN THE REPORT TO THE TEACHER:

Date for 1. Time
26.06.2026

Date for 2. time

Date for 3. Time

TEACHER COMMENTS:

REPORT APPROVED BY TEACHER:

Signature

Date

This is the front page for the reports and has to be attached to the report. The original front page should be attached when a revised report is handed in.

THE BIOLOGY STUDY BOARD

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Cell cycle analysis (2a):

Introduction

In this experiment, flow cytometry-based cell cycle analysis were used to investigate the effects of sodium butyrate (NaBu) on DLD-1 cell lines, for week 2 resveratrol was also used as treatment. Flow cytometry with propidium iodide (PI) staining allows quantification of cellular DNA content and determination of the distribution of cells in the G1, S, and G2/M phases of the cell cycle. Changes in cell cycle distribution can indicate altered proliferation or cell cycle arrest induced by experimental treatments.

Methods

DLD-1 were seeded in T75 flasks and cultured for 24 h before treatment. Cells were exposed to either PBS control, 10 mM NaBu or 20 μ M resveratrol for 48 hours..

Following treatment, cells were harvested by trypsinization, washed in PBS w/o + 5 mM EDTA+ 0.5% BSA (PBS-EB), and fixed in 70% ethanol. Fixed cells were stained with propidium iodide and RNase A before analysis by flow cytometry. Data acquisition and analysis were performed using the free software Floreada.io with cell cycle analysis based on the Watson pragmatic model. Single-cell population was gating on FSC-A versus FSC-H plots. Cell cycle distributions was analysed based on DNA content histograms.

Results week 1

Single-cell populations was gated from FSC-A versus FSC-H. The same gate was applied to all samples to minimize analysis bias, gating is seen in figure 1 for PBS-treatment and figure 2 for NaBu-treatment.

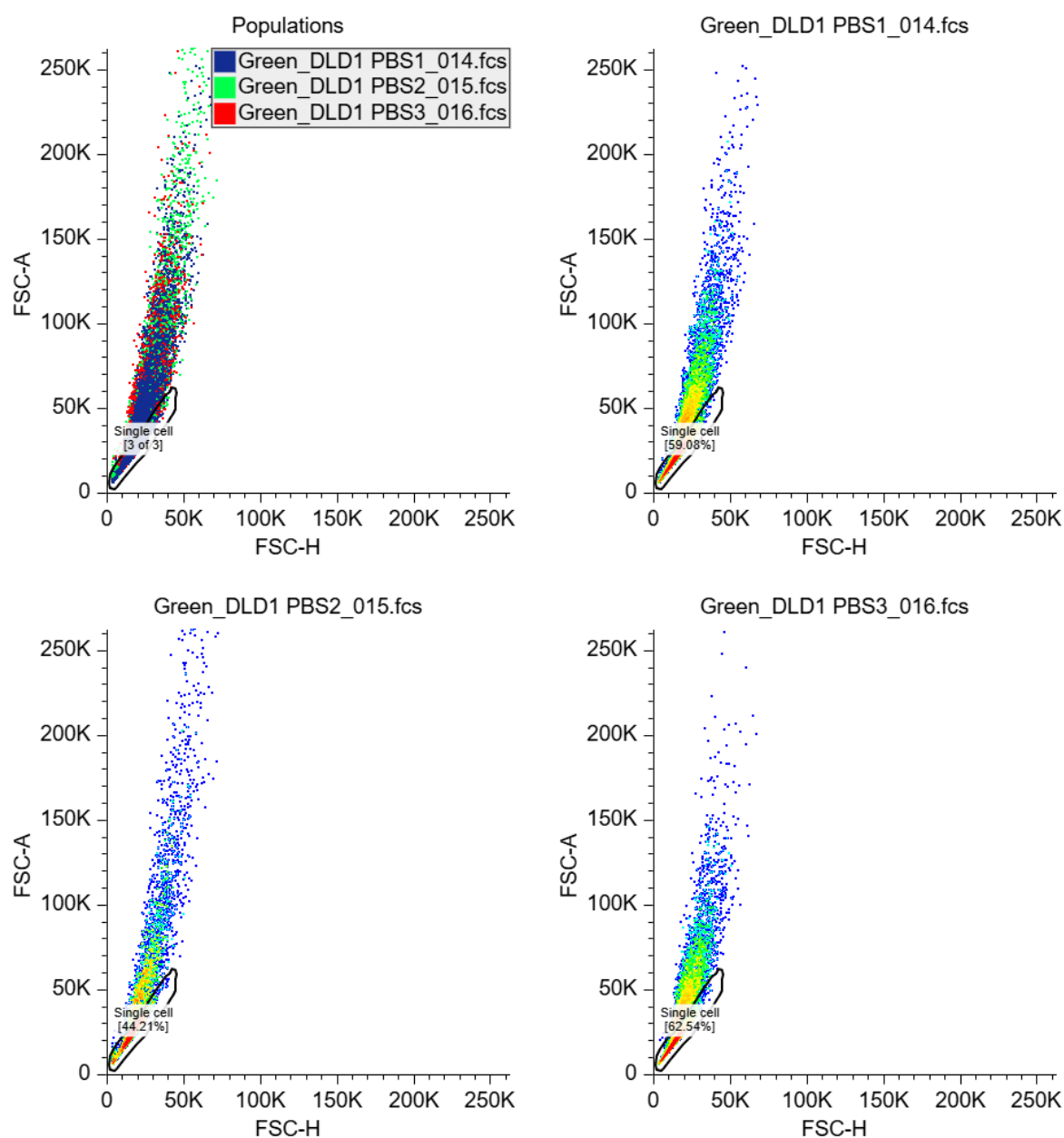


Figure 1. FSC-A plotted against FSC-H for all samples of PBS, and each sample individually. The same gating for single cell, were used for all replicates, to minimize bias. Eventcounts for PBS1= 10.000, PBS2= 3739 PBS2: 10.000 before gating. After gating is applied PBS1= 5908, PBS2= 1653, PBS3= 6254. Data acquisition and analysis were performed using the free software Floreada.io.

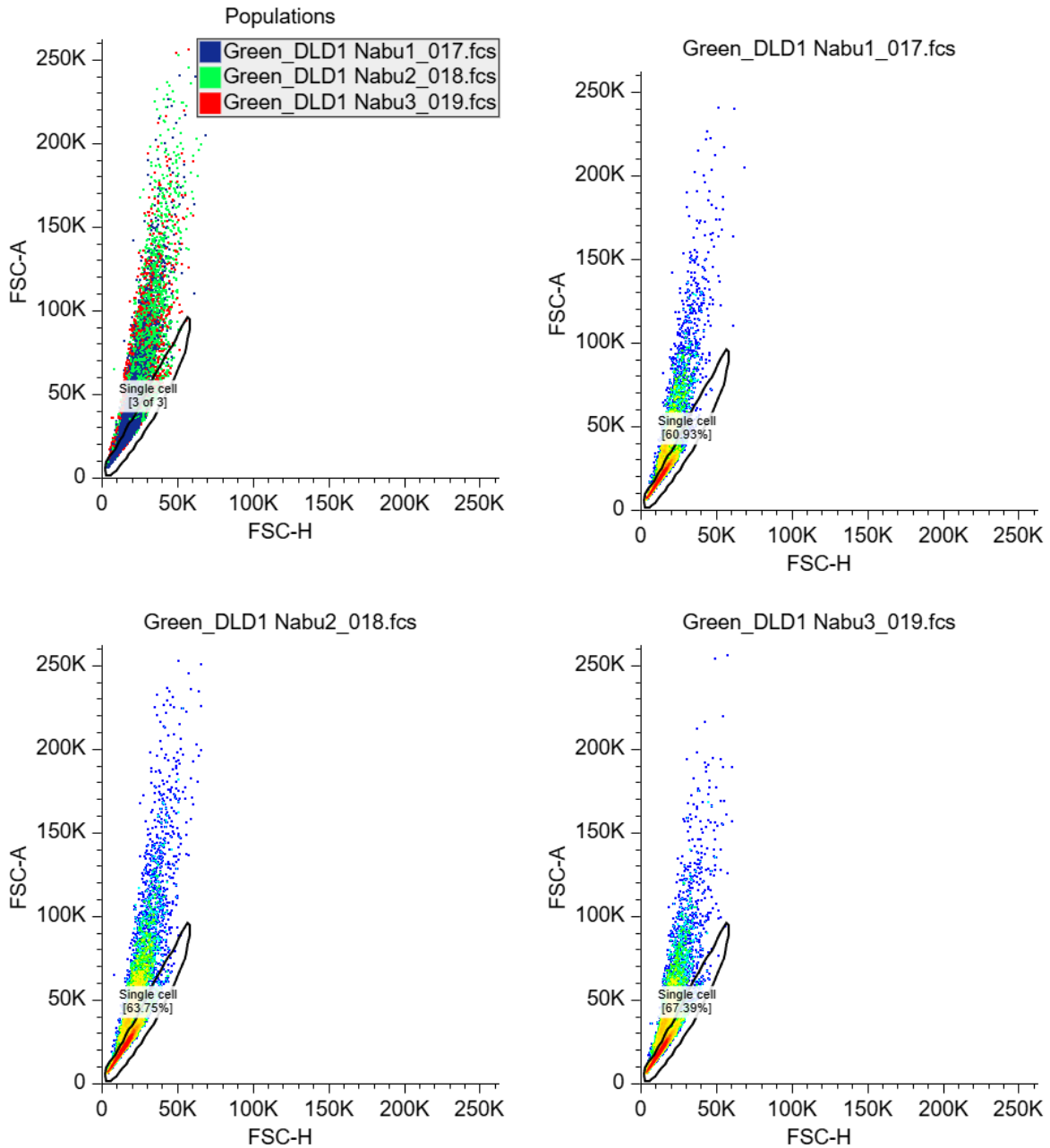
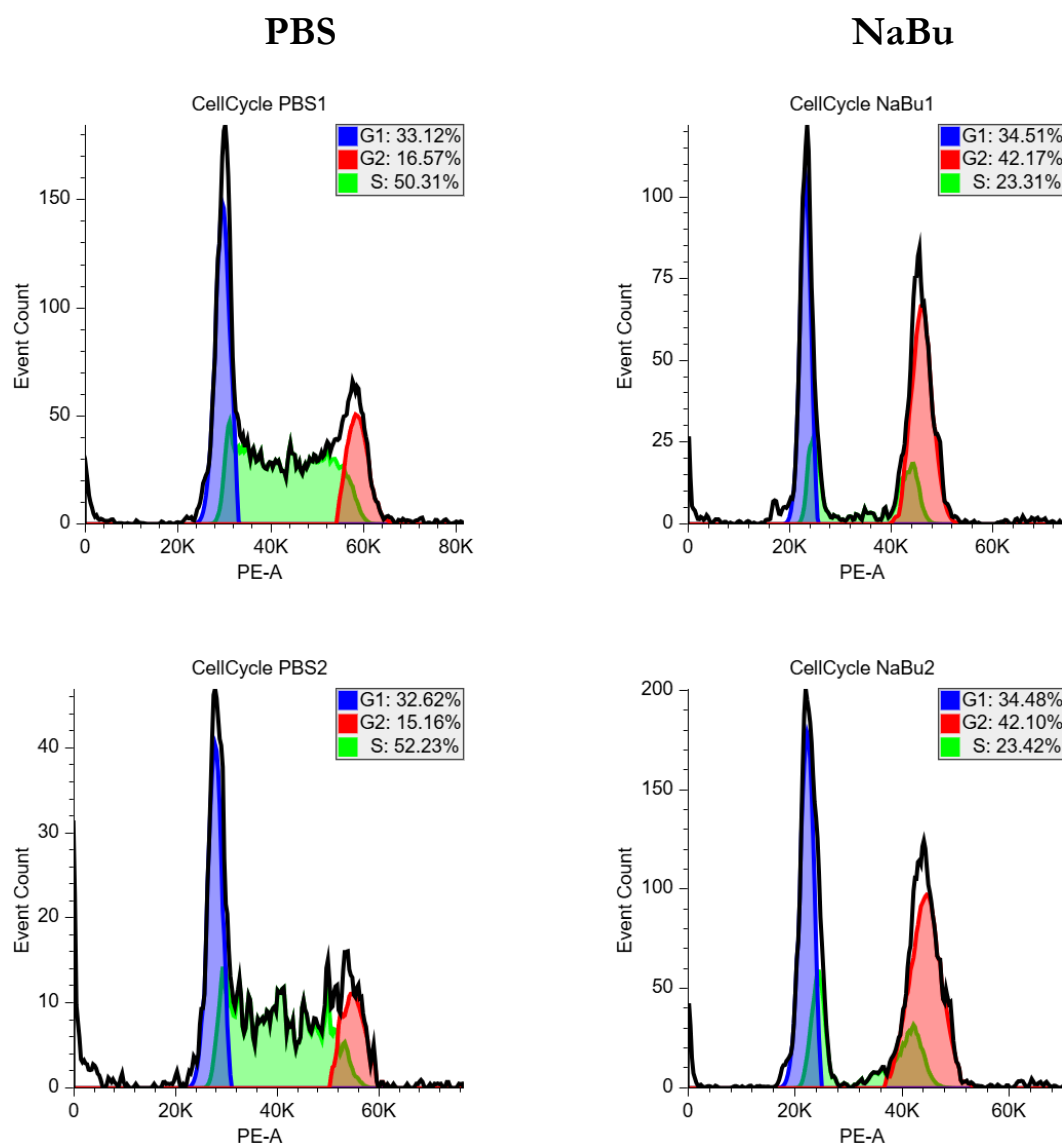


Figure 2. FSC-A plotted against FSC-H for all samples of NaBu, and each sample individually. The same gating for single cell, were used for all replicates, to minimize bias. Eventcounts for NaBu1= 5076, NaBu2: 10.000, NaBu3=10.000 before gating.

After gating is applied NaBu1= 3093, NaBu2= 6375, NaBu3= 6739

Data acquisition and analysis were performed using the free software Floreada.io.

As observed in figure 3, the PBS-treated control cells showed a relatively similar cell cycle distribution across all three replicate. Approximately 33% of cells were present in G1 phase, 50–52% in S phase, and 15–18% in G2 phase. Treatment with NaBu altered DNA content histograms compared with PBS-treated controls as displayed on figure 3. In the NaBu-treated samples, the population in G2/M phase appeared increased relative to the controls. However, the automatic cell-cycle modelling performed by floreada.io did not accurately fit the NaBu histograms, as the fitted S-phase distribution overlapped substantially with the G2/M. The calculated percentage should be regarded as approximate estimates rather than quantitative measurements. However the visual shift of the histograms supports an accumulation of cells in G2/M following NaBu treatment.



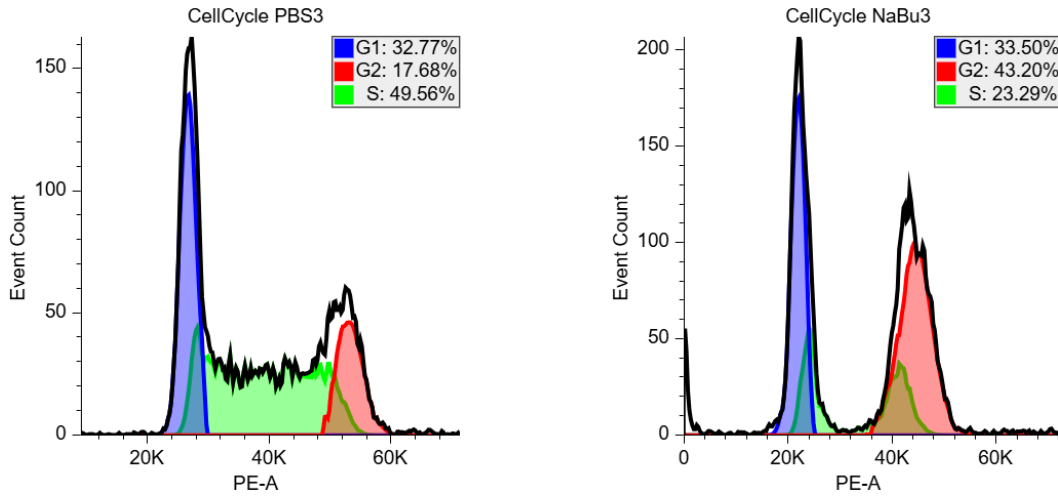


Figure 3. DNA-content histograms of gated single DLD-1 cells after treatment with PBS or 10 mM sodium butyrate (NaBu) for 48 h. Histograms were generated in Floreada.io using the Watson pragmatic model. Although the software estimated the proportion of cells in G1, S and G2/M phases, the automatic fitting did not adequately resolve the histograms because the fitted S-phase distribution overlapped the G2/M peak. Therefore, the displayed percentages should be interpreted as approximate.

	G1	G2	S
PBS1	33,12%	16,57%	50,31%
PBS2	32,62%	15,16%	52,23%
PBS3	32,77%	17,68%	49,56%
	G1	G2	S
NaBu1	34,51%	42,17%	23,31%
NaBu2	34,48%	42,10%	23,42%
NaBu3	33,50%	43,20%	23,29%

Table 1. Estimated cell-cycle phase distribution generated automatically by Floreada.io using the Watson Pragmatic model. Due to imperfect fitting of the Histograms, particularly the overlap between the S-phase and the G2/M-phase, these values should be interpreted only as approximates estimates, rather than precise measurements.

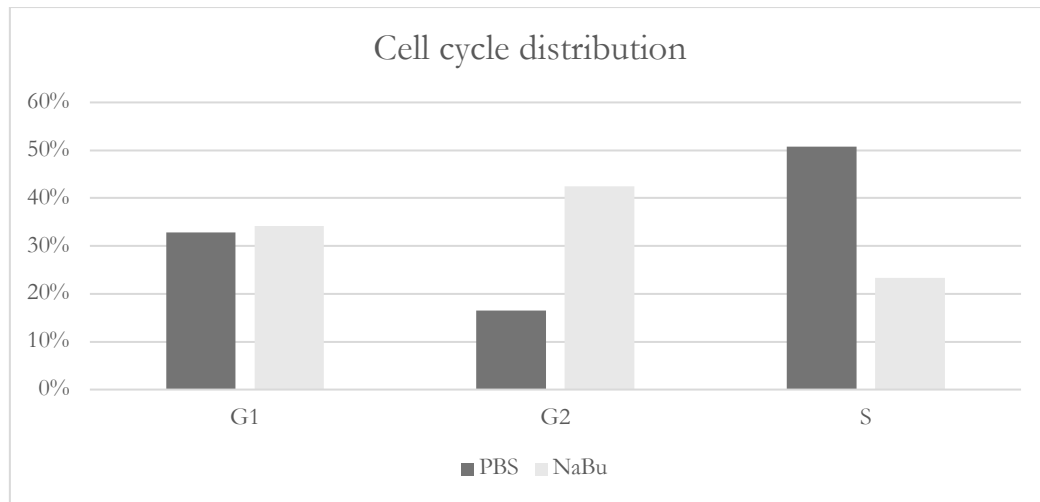


Figure 4. Estimated cell-cycle phase distribution of DLD-1 cells treated with PBS or 10 mM NaBu. Bars represent the mean estimated distribution obtained from Floreada.io. Values should be interpreted qualitatively rather than as exact percentages.

Results week 2

Cell cycle analysis was repeated in week 2 using PBS-treated control cells together with cells treated with 10 mM sodium butyrate (NaBu) or 40 μ M resveratrol as displayed on figure 5.

The PBS-treated cells displayed the expected DNA-content histogram with distinct G1 and G2/M peaks and an intermediate S-phase population. Similar to the week 1 experiment, NaBu treatment visibly altered the DNA-content distribution by increasing the DNA content, consistent with an accumulation in the G2/M phase.

As in the week 1 analysis, the automatic Watson model implemented in Floreada.io did not accurately resolve the histograms. Therefore, the numerical phase percentages were not considered reliable and were not used for quantitative interpretation. Instead, conclusions were based on the qualitative changes observed in the DNA-content histograms.

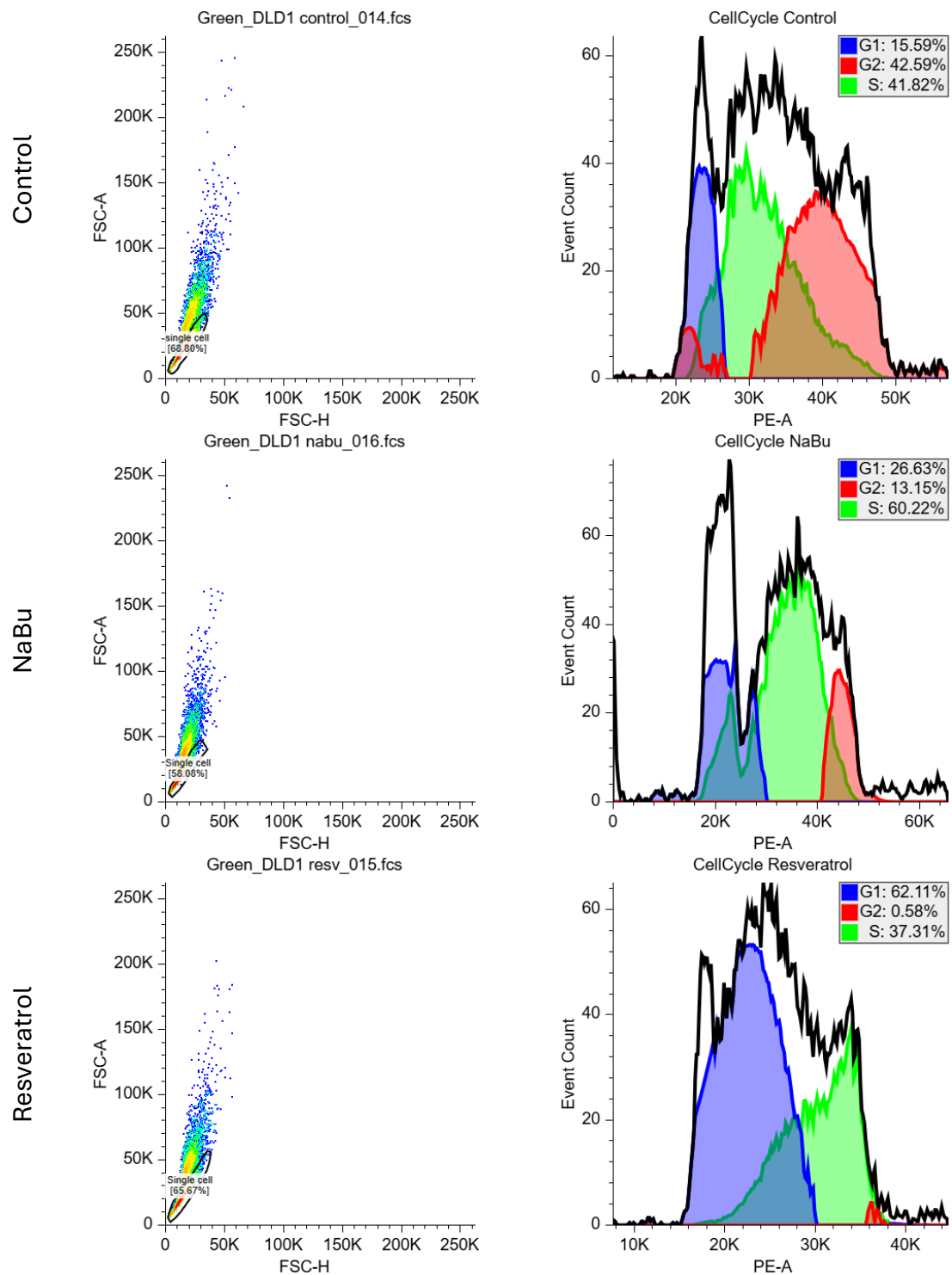


Figure 5. FSC-A plotted against FSC-H and Cell cycle analysis for each sample in week 2 (PBS, NaBu, Resv.). Event counts for all samples before gating was 10,000. Data acquisition and analysis were performed using the free software Floreada.io with Cell Cycle analysis based on the Watson pragmatic model.

Discussion

Cell cycle analysis estimates proliferation by determining the percentage of cells in G0/G1, S, and G2/M phases. This provides information about the distribution of cells across different phases of the cell cycle, and not just whether the cells are proliferating, which is an advantage because it can identify cell cycle arrest or delays caused by exposure of chemicals, drugs etc. Another advantage of the cell cycle analysis is also that it analyses a large number of cells. Some of the drawback from cell cycle analysis is that it measures DNA content rather than DNA synthesis directly, which makes it an indirect measurement of proliferation, and it cannot distinguish between cells that are actively progressing through the cell cycle and cells that are arrested in a cell cycle phase.

It provides a insight into the cell population at one time (snapshot), so if you want to measure at different timepoint, new experiments should be initiated with different time points for harvesting. Furthermore, cells with abnormal DNA content can complicate the interpretation.

Bromodeoxyuridine (BrdU) incorporation directly measures DNA synthesis because BrdU is incorporated into newly replicated DNA during S phase of the cell cycle. Therefore, it could be interpreted as a more direct measurement of proliferating cells. However, BrdU assays require additional staining procedures, DNA denaturation to expose incorporated BrdU, and may affect cell physiology. Unlike cell cycle analysis, BrdU labeling specifically identifies cells actively synthesizing DNA.

Cell cycle analysis provides broader information about cell cycle dynamics and checkpoints, whereas BrdU incorporation offers direct measurement of DNA replication and cell proliferation.

The flow cytometry experiments performed in both weeks consistently showed that NaBu altered the DNA-content distribution of DLD-1 cells. Visual inspection of the histograms demonstrated an increased population of cells with DNA content following NaBu treatment, suggesting an accumulation in the G2/M phase of the cell cycle. Although Floreada generated numerical estimates for the G1, S and G2/M fractions, the automatic fitting was suboptimal for the NaBu-treated samples because the fitted S-phase distribution overlapped with the G2/M peak. Consequently, these percentages should be regarded as approximate estimates rather than precise measurements.

The observed accumulation of cells with DNA content is consistent with a NaBu-induced delay or arrest in G2/M and provides a explanation for the reduced proliferation observed in the Hoechst and resazurin assays (see results from Hoechst and resazurin assay). Cells that fail to progress through mitosis divide less frequently, leading to a reduction in cell number while maintaining metabolic activity proportional to the remaining viable cells. Together, the viability

assays and flow cytometry therefore support the conclusion that NaBu primarily inhibits proliferation by interfering with cell-cycle progression rather than causing acute cell death.

Hoechst 33342 / Alamar Blue 96 well assays (2b):

Introduction

In this experiment, fluorescence-based viability assays were used to investigate the effects of sodium butyrate (NaBu) on DLD-1 and Caco-2 cell lines. Cell viability and metabolic activity were assessed using the Hoechst 33342 and resazurin (Alamar Blue) assays. Hoechst 33342 binds DNA and provides an estimate of relative cell number, whereas resazurin is converted to a fluorescent product by metabolically active cells and reflects cellular metabolic activity.

Methods

Week 1

DLD-1 and Caco-2 cells were seeded in 96-well plates at 10,000 cells per well. Cells were treated with increasing concentrations of NaBu (0.5, 1, and 5 mM). For viability measurements, cells were incubated with Hoechst 33342 (5 µg/mL) and resazurin (25 µg/mL). Resazurin fluorescence was measured first to estimate metabolic activity. Cells were subsequently washed and Hoechst fluorescence was measured to estimate relative cell number. Results were normalized to untreated control cells.

Week 2

We deviated from the protocol in week 2 by accident and only made one plate with DLD-1 occupying columns 2-6 and Caco-2 occupying columns 7-11. Due to this mistake we could not make a dose response curve, but instead we went with a binary assay using fixed concentrations of 1 mM NaBu and 40 µM Resveratrol, both alone and combined. Cells were incubated with the same Hoechst and resazurin concentrations as in week 1 and fluorescence was measured. Results were normalized to untreated control cells. Note that since columns 2 and 11 in the plate contained cells+media (but not media+vehicle) they were not used for the final analysis (see page the appendix, page 36).

Results

Week 1

The effect of NaBu on cell number was assessed using the Hoechst 33342 assay, while cellular metabolic activity was evaluated using the resazurin assay.

For DLD-1 cell line NaBu treatment reduced both cell number and metabolic activity in a concentration-dependent manner as seen on figure 5. Relative cell number measured by Hoechst fluorescence decreased from 100% in the control group to approximately 88%, 67%, and 26% following treatment with 0.5, 1, and 5 mM NaBu, respectively.

Similarly, resazurin fluorescence decreased to approximately 87%, 65%, and 23% of control values at 0.5, 1, and 5 mM NaBu, respectively. The reduction in metabolic activity closely followed the reduction in cell number.

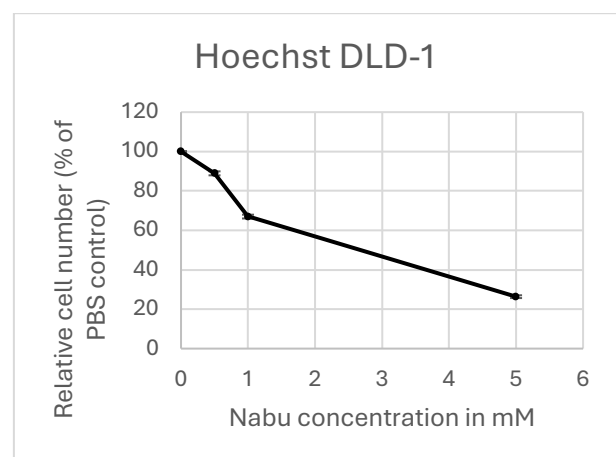
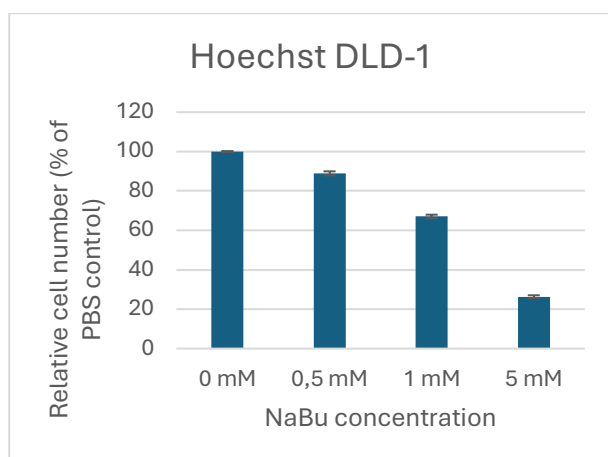
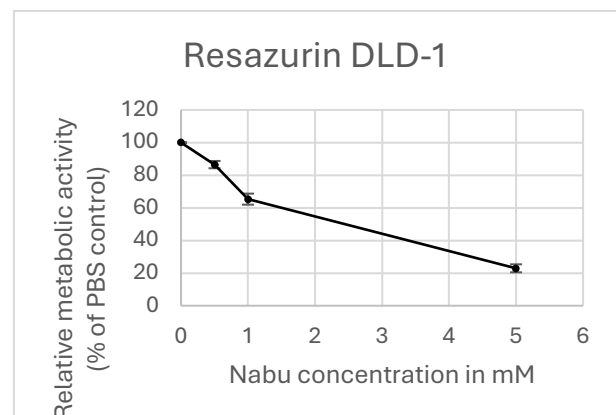
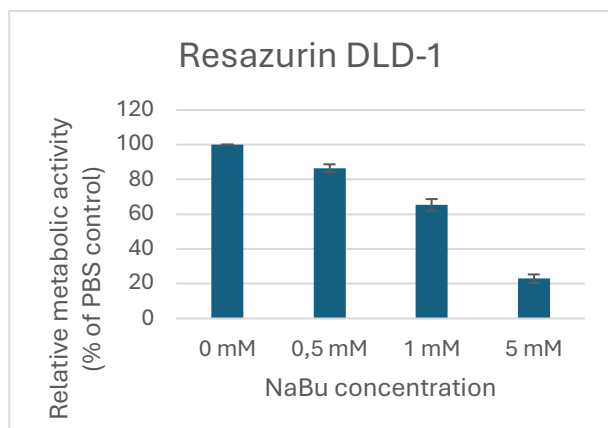


Figure 6. Effect of increasing NaBu concentrations (0, 0.5, 1 and 5 mM) on DLD-1 cells measured using resazurin (metabolic activity) and Hoechst 33342 (relative cell number). Fluorescence values were normalized to untreated controls (100%). Bars represent the mean of replicate measurements, and error bars indicate $\pm$ SEM.

Caco-2 cells appeared less sensitive to NaBu treatment than DLD-1 cells. At 0.5 and 1 mM NaBu, Hoechst and resazurin signals remained close to control levels. At 5 mM NaBu, however, cell number decreased to approximately 61% of control and metabolic activity decreased to approximately 57% of control.

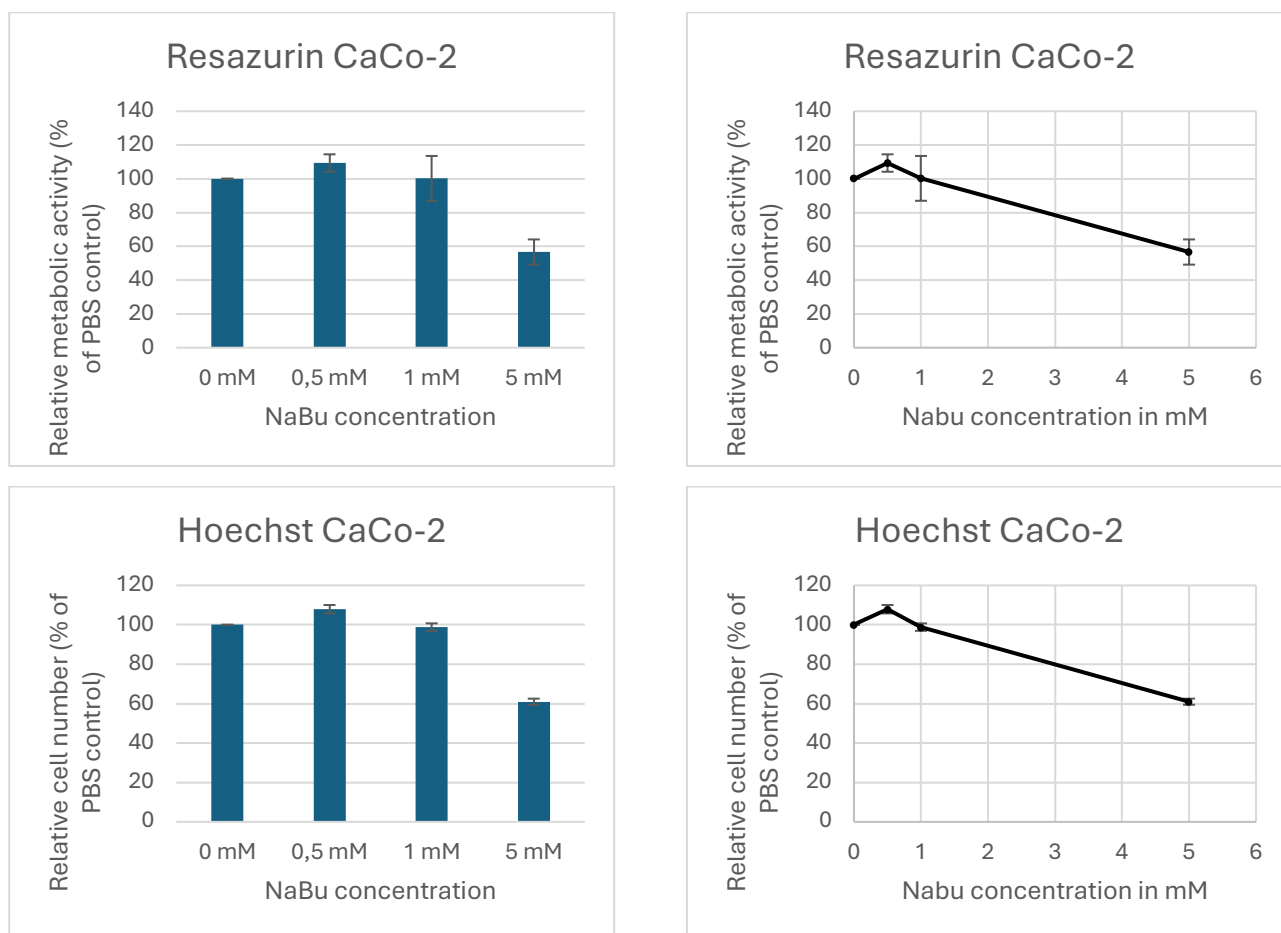


Figure 7. Effect of increasing NaBu concentrations (0, 0.5, 1 and 5 mM) on Caco-2 cells measured using resazurin (metabolic activity) and Hoechst 33342 (relative cell number). Fluorescence values were normalized to untreated controls (100%). Bars represent the mean of replicate measurements, and error bars indicate $\pm$ SEM.

Week 2:

The week 2 experiment reinforced the previous NaBu results, with DLD-1 cells responding relatively strongly to 1 mM NaBu by showing a ~30-40% decrease in fluorescence intensity relative to the control for both stains (figure 2). Addition of 40 μ M Resveratrol strongly reduces cell proliferation by a factor of ~x10, and the combination of NaBu and Resveratrol shows the same result. For Caco-2 cells, we observe an increase in cell proliferation for both NaBu and Resveratrol treatment, and only the combined treatment shows a level around the control. However, we must note that our Caco-2 samples for week 2 showed a very high variance for multiple treatments, making it harder to draw firm conclusions. In addition, while the relative signal for the two stains is quite aligned for DLD-1 cells it is more decoupled for Caco-2, with the Hoechst signal showing a greater relative effect for all treatments.

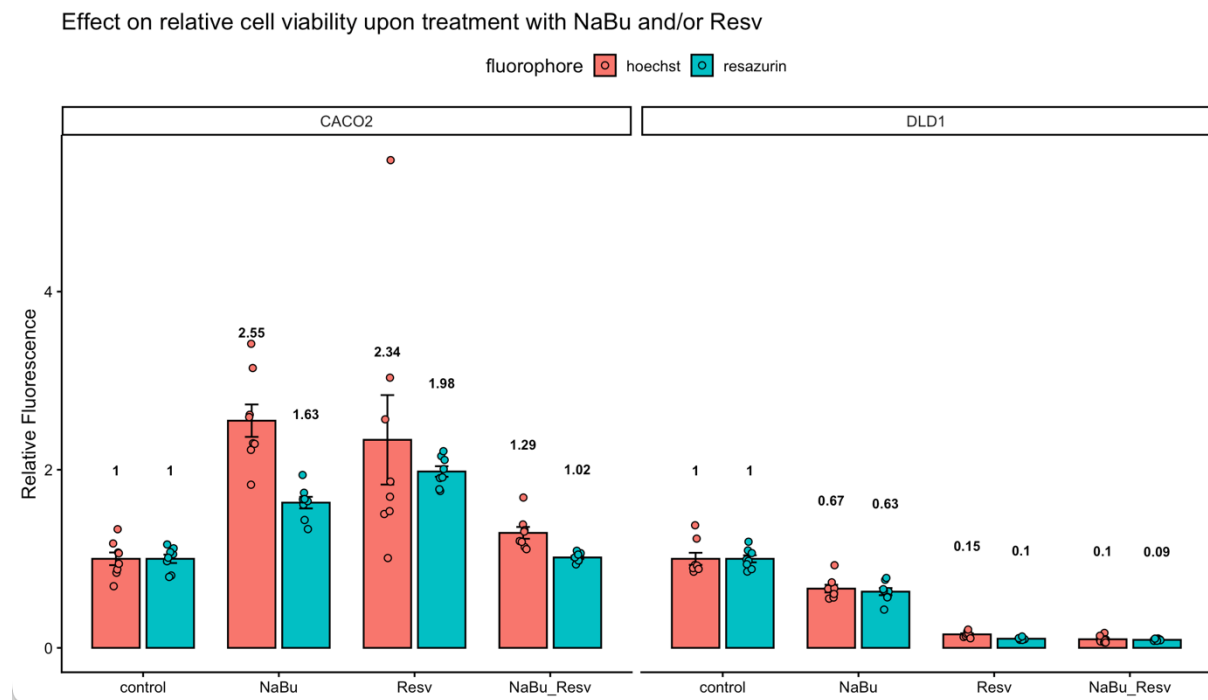


Figure 8, Cell viability measured both by DNA synthesis (Hoechst stain) and cell metabolism (resazurin stain).

The control samples used vehicle PBS and DMSO in matching volumes. Concentration of NaBu: 1mM, concentration of Resveratrol: 40 μ M. The bar charts indicate the group means relative to the control. Error bars: standard deviation of the mean.

Discussion

In the week 1 assay, the Hoechst 33342 stain demonstrated that sodium butyrate reduced the relative number of viable cells in both cell lines. However, the effect was substantially greater in DLD-1 cells than in Caco-2 cells. In DLD-1 cells, cell number decreased progressively with increasing NaBu concentration, reaching approximately 26% of control at 5 mM NaBu. In contrast, Caco-2 cells were less affected at lower concentrations and retained approximately 61% of control cell number at 5 mM NaBu. The resazurin assay showed a similar concentration-dependent decrease in metabolic activity. The decrease in resazurin fluorescence closely matched the decrease in Hoechst fluorescence.

For the week 2 assay, we again saw more evidence that DLD-1 cells responded to 1mM NaBu by having reduced proliferation, as evidenced by both Hoechst and resazurin fluorescence being decreased in tandem, while Caco-2 cells in fact showed increased proliferation in our assay. DLD-1 cells showed a very strong response to 40 μ M Resveratrol and Resveratrol+NaBu. Due to the uncertainty in our results, especially for the Caco-2 cells, more research needs to be conducted to verify whether this cell line indeed shows a replicable increase in cell proliferation in response to NaBu or Resveratrol.

The flow cytometry results indicated that the NaBu treatment caused an accumulation in the proportion of cells in the G2 phase of the cell cycle and a corresponding decrease in the S-phase population. This indicates that NaBu induced cell cycle arrest, which likely contributed to the reduction in cell proliferation observed in the viability assays.

Conclusion

Treatment with NaBu reduced both the relative cell number and metabolic activity of DLD-1 cells in a concentration-dependent manner, whereas Caco-2 cells were less sensitive to the treatment. Resveratrol causes a strong reduction in the overall relative cell number and metabolic activity in DLD-1 cells, while leaving Caco-2 cells more or less unaffected, although the uncertainty for the Caco-2 results warrant further investigation.

Flow cytometry demonstrated that NaBu altered the DNA-content distribution of DLD-1 cells, consistent with an accumulation of DNA content and therefore indicative of cell-cycle arrest. Because the automatic cell-cycle fitting performed by Floreada.io inadequately histograms, the

calculated phase percentages should be interpreted cautiously. Nevertheless, the qualitative changes observed in the DNA-content histograms together with the viability assays support the conclusion that NaBu suppresses DLD-1 cell proliferation.

Appendix:

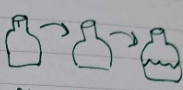
Lab journal:

2A:

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pre-trypsin
cell phenotype: red medium
~ 50% confluence
small clusters of cells

3x T75 flasks switched & washed x2 w/ Na citrate



After trypsinization we collected all cell suspension in 1xT75 flask and mechanically splitted them before we transferred 200 µl in saline for cell count on the coulter counter. We looked in the microscope to check the cells were single celled.

Coulter 1: 11778 (0.5 ml sucked, 200 µl in 10 ml saline → x 702 factor)

Coulter 2: 11995

avg: 11886.51.2 total 1.212'372 cells/µl

passage $300'000 / (1.212'372 \text{ cells}/\mu\text{l}) = 0.247$

we take 250 µl → 303.095 cells

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Side Side 6

Caco2	
Wash 2 times with 10ml 85 mM sodium citrate	
Trypsinate with 1ml trypsin-EDTA	
Split cells	
Resuspend in 10 ml media	
Make new T75 for other days	
0.2-0.5 million cells pr passage	
Make mastermix for experiment 1,3 and 4	
500.000 cells + 500.000 cells + 1.500.000 cells + 250.000 cell = 2.750.000 cells (20% extra is included)	
0.2-0.5 million cells pr passage	
Use C1*V1=C2*C2 to calculate correct amounts of cell-suspension for mastermix	

exp. 2A: DLD-1

- we take two bottles and put 0.21 mL McCoy in each as well as 0.77 mL DLD-1, with so we can have 200,000 cells in each

exp. 2B: DLD-1

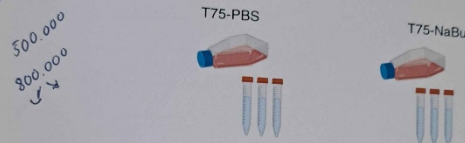
- protocol says to make a master mix of 10 mL, in order to use the multipipet so we will double it to 20 mL by doubling $9.44 \approx 18.88$ mL McCoy and $0.56 \times 2 = 1.12$ mL

- we ^{put} take μ L of the master mix per well and add 180 μ L McCoy per well and put the well plates in heat

17.06.2026 green team: Line, Lars, Andreas
 DLD-1:
 Experiment 2a:
 Cell cycle analysis

Preparation of cell samples for Flow cytometry.

- Trypsinize cells. Resuspend cells in PBS-EB (PBS w/o + 5 mM EDTA + 0.5% BSA) and pellet cells by centrifugation for 5 min at appr. 300g, about 1800 rpm. (Note: If you want to analyze apoptotic cells as well, remember to collect the cells floating in the media—but we do not include these cells for this experiment). Discard supernatant.
- Wash cells in appr. 10 mL PBS-EB
- Spin down as before. Discard supernatant. Resuspend cells in 3-5 mL PBS-EB.
- Count the cells. Transfer 500,000 cells to a FACS tube (BD, Falcon 5 mL polystyrene round-bottom tube, cat # 35 2058). Add carefully PBS-EB until the total volume is appr. 3 mL. Spin down the cells and discard supernatant. Make up to 3 of these FACS tubes for each cell line.



Resuspend cells in exactly 200 μ L PBS-EB. Add slowly 2 mL ice-cold 70% ethanol while **whirly mixing (IMPORTANT)**. Incubate on ice for at least 30 min. ~~The samples should be kept at -20°C at this stage.~~

Staining the cells for Flow cytometry

- Centrifuge the cells at 300g and 4°C for 5 min. Discard supernatant.
- Resuspend cells in 2-3 mL PBSw. Spin as before. Discard supernatant.
- Resuspend cells in 400 μ L PBSw. Add 50 μ L RNase (1 mg/mL) and 50 μ L propidium iodide (0.4 mg/mL). Incubate at RT for 30 min in the dark. 15.05
- Centrifuge the cells at 300g and 4°C for 5 min. Discard supernatant.
- Resuspend cells in 0,5 mL PBSw
- Analyze by flow cytometry.

do not centrifuge (that doesn't cool)
 - Washed with PBS 1/2 first 10 mL
 - we take our cells from incubation and trypsinize them. we accidentally washed the NaBu samples.

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27-06-2026

taken out samples from the 37°C incubator, sucked the media from the T75 Flask and added 1ml Trypsin in both. Next we incubated the cells with Trypsin in 37°C for 6min. We looked in the microscope to confirm de-attachment and then resuspended the cells in 13ml PBS w/o + EDTA + BSA for both T75-Flasks

10ml PBS w/o ++

10ml PBS w/o ++

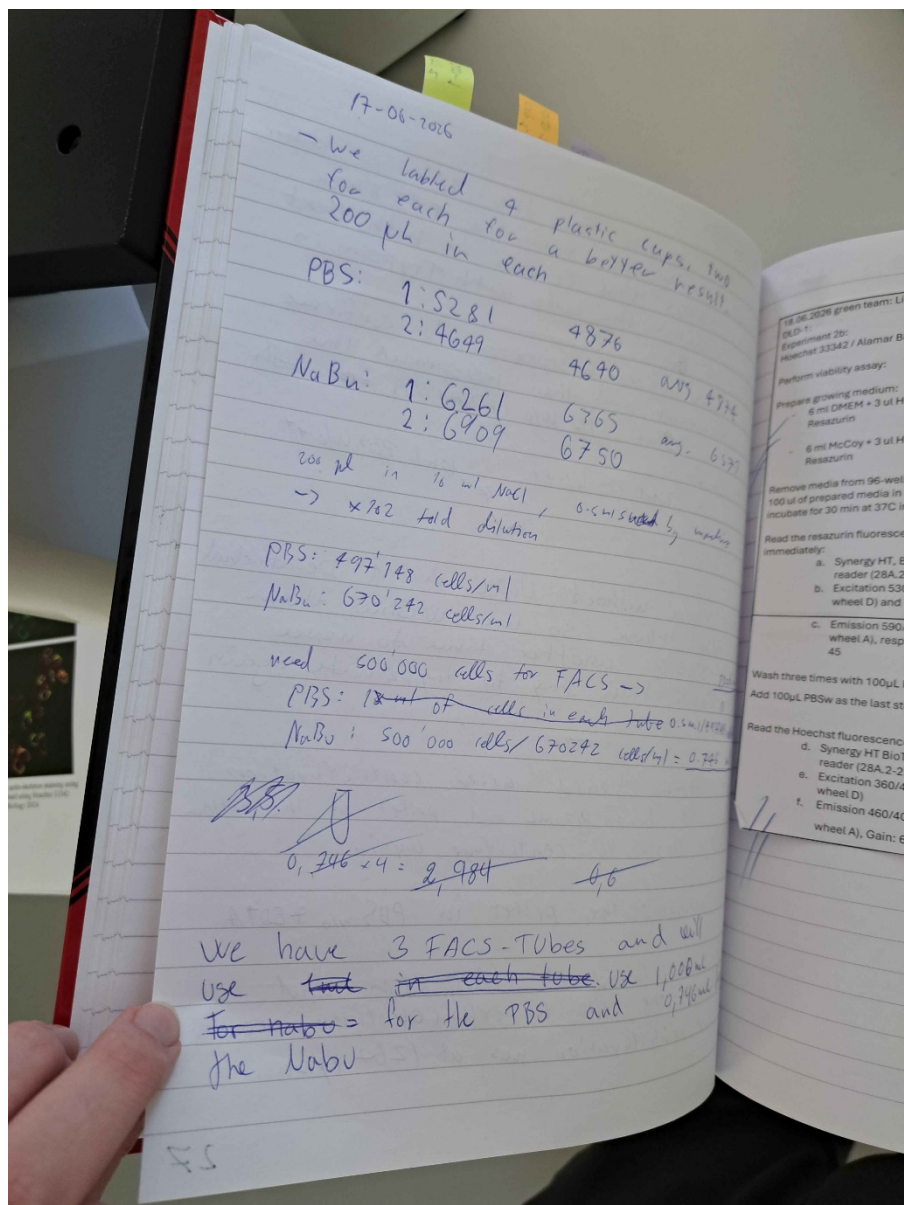
the Flask without NaBU didn't de-attach from the Flask so we washed them with PBS % another time to make sure and then try to trypsinise again

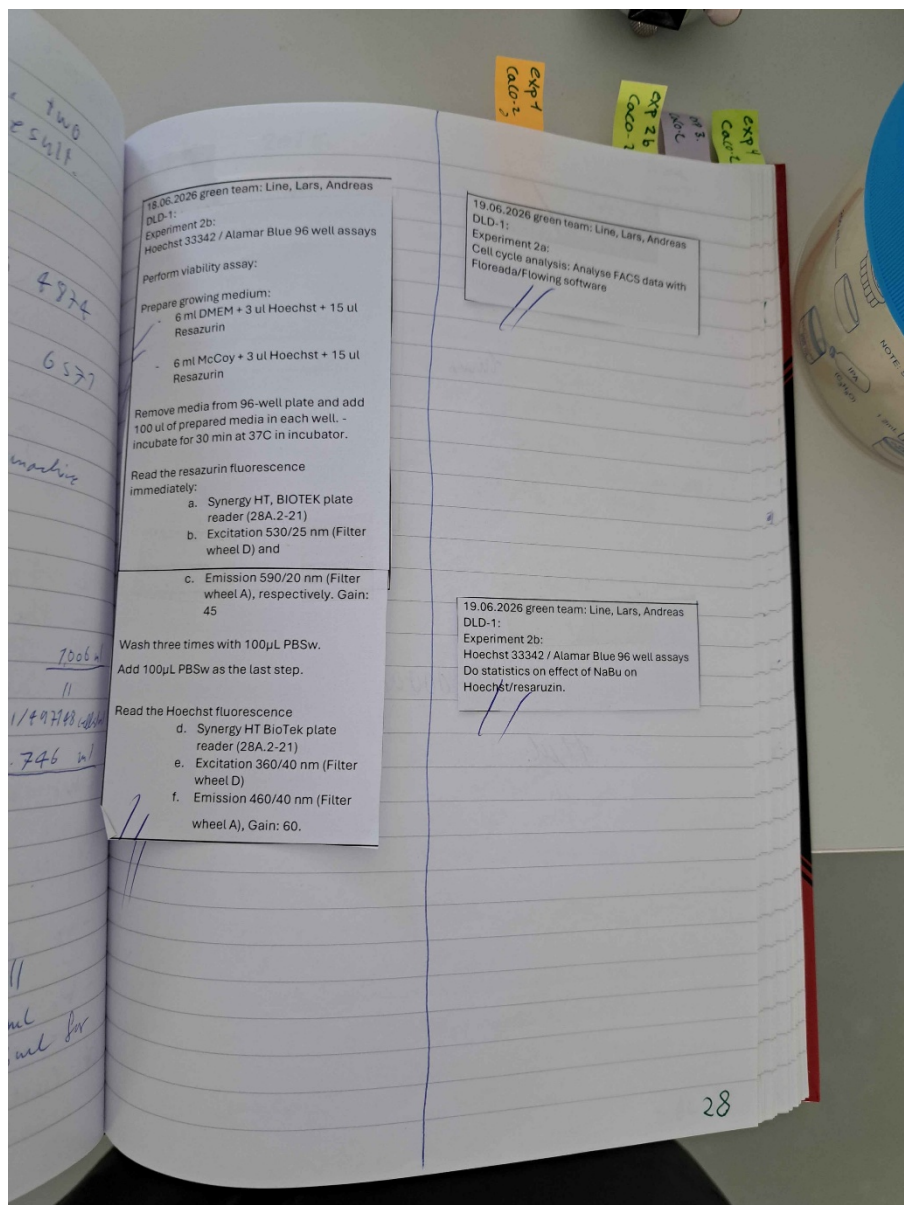
- we remove liquid after centrifugation until we have 0.5 ml and pellet at the bottom of the flask to centrifuge again.

- we resuspend the pellet in PBS w/o + EDTA

- we centrifuged at 1800 RPM and prepared the cells for cell counting; counter

Obs: First centrifugation was at 1264 rpm





22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Seed for experimental 2A: cell cycle analysis

Seed 3 x T75 flask with 700.000 cells in each flask, total of 12 ml/flask.

$$\frac{700.000}{1.131.180} = 0,618 \text{ ml} + 12 \text{ ml media}$$

cell suspension

1153 and 10649 cells pr
0,5 ml from coulter counter
 $\frac{1153 + 10649}{2} = 11.090 \cdot 59 \cdot 2 = 1.131.180$
cells/ml.

22.06.2026 green team: Line, Lars, Andreas
DLD-1:

Experiment 2b:

Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A	No cells - media only	Control	1 mM NaBtu + DMSO	5 mM NaBtu + DMSO	PBS + 10 μ M Rev	1 mM NaBtu + 10 μ M Rev	5 mM NaBtu + 10 μ M Rev	5 mM NaBtu + 20 μ M Rev	PBS + 40 μ M Rev	1 mM NaBtu + 40 μ M Rev	5 mM NaBtu + 40 μ M Rev	No cells - media only
B												
C												
D												
E												
F												
G												
H												

Figure 2.2

Seed one 96 well plate with 10.000 DLD-1 cells pr well. Volume of 200 μ L pr well.

Mastermix:

10ml with 500.000 cells

$$\frac{500.000}{1.131.180} = 0,442 \text{ cell sop} + 9,958 \text{ Media}$$

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23.06.2026 green team: Line, Lars, Andreas
DLD-1
Experiment 2a: cell cycle
Treat with PBSw, 20 μ M Resv and 10 mM
NaBu

23-06-2026 green team: Line, Lars,
Andreas
DLD-1: McCoy medium
Experiment 2A:
Need PBSw, 20 μ M Resv and 10 mM NaBu
75 μ l PBS + 3 μ l DMSO in 15 ml media
75 μ l PBS + 3 μ l Resveratrol in 15 ml media
Use 75 μ l NaBu 3 μ l DMSO in 15 ml media

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24.06.2026 green team: Line, Lars, Andreas
DLD-1

Experiment 2a: cell cycle

FACS

Input: 3x treated T75 flasks

Workflow:

- Trypsinize
- Resuspend in 9 ml PBS-EB
- Centrifugate for 5 min at 300g then discard supernatant
- Wash in 10 ml PBS-EB
- Centrifugate 5 min at 300g then n discard supernatant
- Resuspend in 3 ml PBS-EB
- Count cells
- Transfer 500'000 cells to FACS tube fill until 3 ml.
- Generate more tubes if enuff cells
- Centrifugate and discard supernatant
- Resuspend in 200 μ l PBS-EB
- Add 2 ml ice-cold 70% ethanol while whirly mixing (vortex).
- Incubate on ice for at least 30 minutes. Keep samples on -20.
- ---- staining step -----

- Centrifugate at 300g for 5 min. Discard supernatant.
- Resuspend in 2-3ml PBSw.
- Centrifugate at 300g for 5 min. Discard supernatant
- Resuspend in 400 μ l PBSw. Add 50 μ l RNAse and 50 μ l propidium iodide (use gloves). Incubate at room temperature for 30 min.
- Centrifugate at 300g for 5 min. Discard supernatant.
- Resuspend in 0.5 ml PBSw.
- Analyze by flow cytometry

confluences:

ctrl: 40%.

Resv: 90%.

NaBu: 38%.

not enough, 40 with 9 ml

with PBS-EB

slowly

cells/ml
concentration

counts: ctrl 1: 673 62425

200 μ l ctrl 2: 698 71196

x902 NaBu 1: 793 80886

NaBu 2: 897 90882

Resv 1: 447 45594

Resv 2: 535 54570

volumes after (looking at 15 μ l tube)

ctrl: 2.5 ml

Resv: 2.5 ml

NaBu: 2.6 ml

FACS seems fine (number of cells)

69

Experiment 2B:

15. Juni 2020 grön gruppe

EXP 2 DLD-1

Mastermix	Seed for experimental 2A: cell cycle analysis
24 ml with 700.000 cells	2 x T75 flask 12 ml with 700.000 cells
Mastermix	Seed for experimental 2B: Hoechst plate reader assay
10ml with 500.000 cells	10.000 cells pr well in 200 μ L
A-H= 800 μ L With 10.000 cells	Seed column 2-6 of a 96 well plate with DLD-1 cells and column 7-11 with Caco-2 cells: 10.000 cells per well. Seed in a volume of 200 μ L. 200 μ L medium without cells is added to column 1 and 12.

DLD-1 2A
calc: 885.870 cells/mL

700.000 mL
885.870 cells/mL = 0,79 mL

0,79 mL DLD-1 2A - 0,79 = 23,21 mL

0,769 mL DLD-1 + 23,21 mL McCoy = 24

Caco-2

11,21 mL medium

0,79 mL cell suspension

DLD-1 2B calc: 885.870 cells/mL

$\frac{500.000}{885.870} = 0,56$ mL 10 mL - 0,56 mL = 9,44 mL

DLD-1: 0,56 mL + McCoy 9,44 mL

Seed cells per well: $\frac{10.000}{885.870 \text{ cells/mL}} = 0,011$ mL ≈ 11 μ L

200 μ L - 11 μ L = 189 μ L

11. $\frac{10.000}{500.000} = 0,02$ mL ≈ 20 μ L DLD + Master mix

200 μ L - 20 μ L = 180 μ L McCoy

200 μ L - 20 μ L = 180 μ L McCoy

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exp. 2A: DLD-1

- we take two bottles and put 0.21 mL McCoy in each as well as 0.77 mL DLD-1, with so we can have 200,000 cells in each

exp. 2B: DLD-1

- protocol says to make a master mix of 10 mL, in order to use the multipipet so we will double it to 20 mL by doubling $9.44 \approx 18.88$ mL McCoy and $0.56 \times 2 = 1.12$ mL

- we ^{put} take μ L of the master mix per well and add 180 μ L McCoy per well and put the well plates in heat

16.06.2026 green team: Line, Lars, Andreas
 DLD-1:
 Experiment 2a: Cell cycle analysis
 NaBu-treated flask (10 mM final)
 • 120 μ L of 1 M NaBu stock
 • 11,88 mL medium
 Total = 12 mL
 PBS vehicle control flask
 To keep the PBS concentration identical to the treated flask:
 • 120 μ L PBSw
 • 11,88 mL medium
 Total = 12 mL

16.06.2026 green team: Line, Lars, Andreas
 DLD-1:
 Experiment 2b: Hoechst 33342 / Alamar Blue
 96 well assays

Prepare 2 mL of each condition

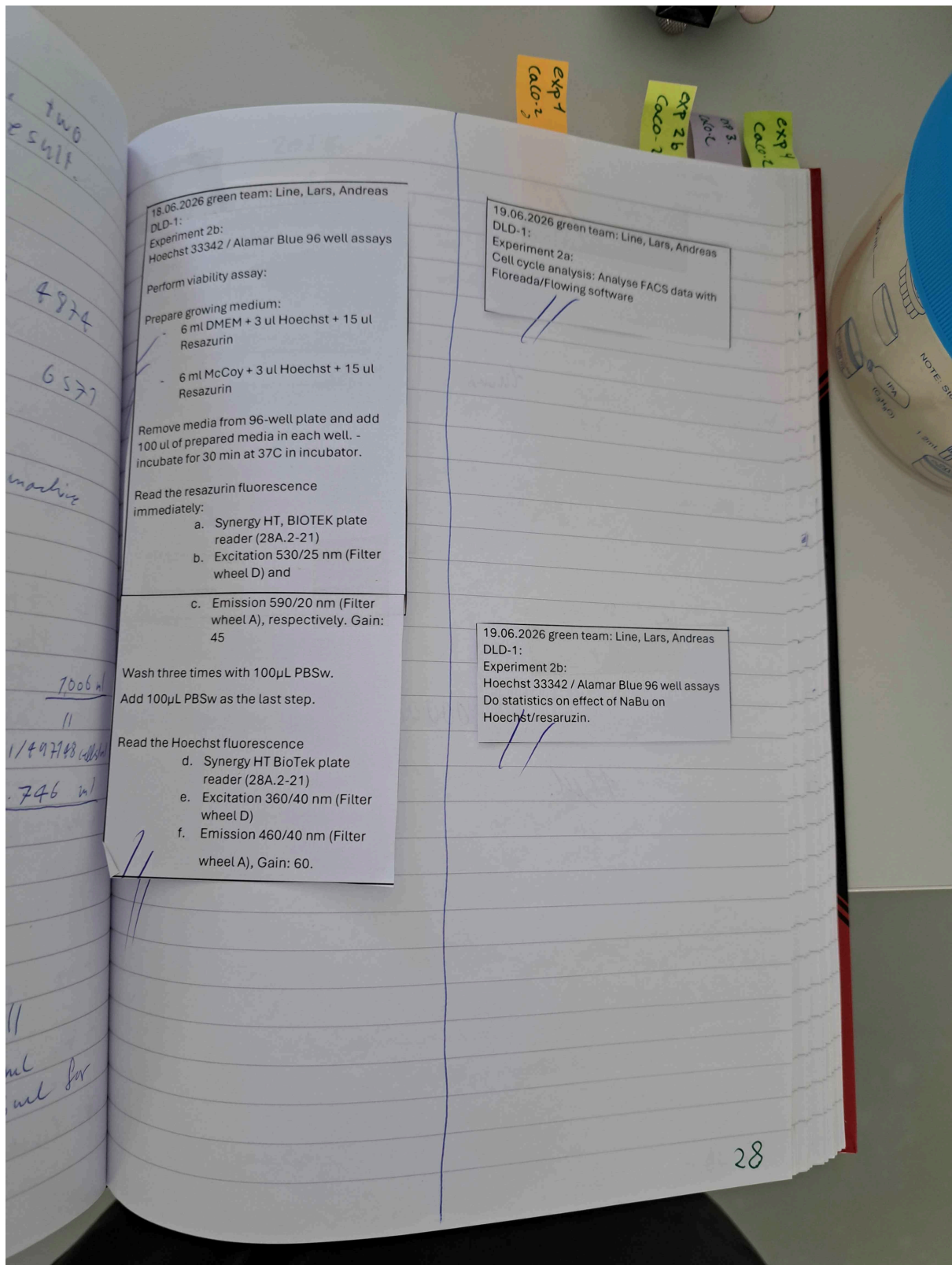
Final NaBu	100 mM NaBu stock	Additional PBSw	Medium	Total
Control (0 mM)	0 μ L	100 μ L	1900 μ L	2000 μ L
0.5 mM	10 μ L	90 μ L	1900 μ L	2000 μ L
1 mM	20 μ L	80 μ L	1900 μ L	2000 μ L
5 mM	100 μ L	0 μ L	1900 μ L	2000 μ L

Well	Condition	Cell Type
A1	Control (0 mM)	DLD-1 cells
B1	Control (0 mM)	DLD-1 cells
C1	Control (0 mM)	DLD-1 cells
D1	Control (0 mM)	DLD-1 cells
E1	Control (0 mM)	DLD-1 cells
F1	Control (0 mM)	DLD-1 cells
A2	0.5 mM	DLD-1 cells
B2	0.5 mM	DLD-1 cells
C2	0.5 mM	DLD-1 cells
D2	0.5 mM	DLD-1 cells
E2	0.5 mM	DLD-1 cells
F2	0.5 mM	DLD-1 cells
A3	1 mM	DLD-1 cells
B3	1 mM	DLD-1 cells
C3	1 mM	DLD-1 cells
D3	1 mM	DLD-1 cells
E3	1 mM	DLD-1 cells
F3	1 mM	DLD-1 cells
A4	5 mM	DLD-1 cells
B4	5 mM	DLD-1 cells
C4	5 mM	DLD-1 cells
D4	5 mM	DLD-1 cells
E4	5 mM	DLD-1 cells
F4	5 mM	DLD-1 cells
A5	Control (0 mM)	Caco-2 cells
B5	Control (0 mM)	Caco-2 cells
C5	Control (0 mM)	Caco-2 cells
D5	Control (0 mM)	Caco-2 cells
E5	Control (0 mM)	Caco-2 cells
F5	Control (0 mM)	Caco-2 cells
A6	0.5 mM	Caco-2 cells
B6	0.5 mM	Caco-2 cells
C6	0.5 mM	Caco-2 cells
D6	0.5 mM	Caco-2 cells
E6	0.5 mM	Caco-2 cells
F6	0.5 mM	Caco-2 cells
A7	1 mM	Caco-2 cells
B7	1 mM	Caco-2 cells
C7	1 mM	Caco-2 cells
D7	1 mM	Caco-2 cells
E7	1 mM	Caco-2 cells
F7	1 mM	Caco-2 cells
A8	5 mM	Caco-2 cells
B8	5 mM	Caco-2 cells
C8	5 mM	Caco-2 cells
D8	5 mM	Caco-2 cells
E8	5 mM	Caco-2 cells
F8	5 mM	Caco-2 cells

Obs: The wells show different colours depending on either DLD-1 or Caco-2

- Old medium is removed and medium with NaBu is added. 2 and 3 with PBS control, and 4 with 0.5 mM NaBu, 5 with 1 mM NaBu, 6 with 5 mM NaBu

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16.06.2026 green team: Line, Lars, Andreas
 CaCo-2
 Experiment 2b: Hoechst 33342 / Alamar
 Blue 96 well assays

Prepare 2 mL of each condition

Final NaBu	100 mM NaBu stock	Additional PBS	Medium	Total
Control (0 mM)	0 μ L	100 μ L	1900 μ L	2000 μ L
0.5 mM	10 μ L	90 μ L	1900 μ L	2000 μ L
1 mM	20 μ L	80 μ L	1900 μ L	2000 μ L
5 mM	100 μ L	0 μ L	1900 μ L	2000 μ L

	1	2	3	4	5	6	7	8	9	10	11	12
A	No cells - media only	Control - PBS	Control - PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	Control - PBS	Control - PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	No cells - media only
B												
C												
D												
E	No cells - media only	Control - PBS	Control - PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	Control - PBS	Control - PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	No cells - media only
F												
G												
H												

Figure 2.1

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exp 1
CaCo-2

18.06.2026 green team: Line, Lars, Andreas
CaCo-2:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. -incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

- g. Synergy HT, BIOTEK plate reader (28A.2-21)
- h. Excitation 530/25 nm (Filter wheel D) and
- i. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

- j. Synergy HT BioTek plate reader (28A.2-21)
- k. Excitation 360/40 nm (Filter wheel D)
- l. Emission 460/40 nm (Filter wheel A), Gain: 60.

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22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Seed for experimental 2A: cell cycle analysis

Seed 3 x T75 flask with 700.000 cells in each flask, total of 12 ml/flask.

$$\frac{700.000}{1.131.180} = 0,618 \text{ ml} + 12 \text{ ml media}$$

cell suspension

1153 and 10649 cells pr
0,5 ml from coulter counter
 $\frac{1153 + 10649}{2} = 11.090 \cdot 59 \cdot 2 = 1.131.180$
cells/ml.

22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A	No cells - media only	Control PBS-DMSO	1 mM NaBu + DMSO	5 mM NaBu + DMSO	PBS + 10 μ M Rev	1 mM NaBu + 10 μ M Rev	5 mM NaBu + 10 μ M Rev	5 mM NaBu + 20 μ M Rev	PBS + 40 μ M Rev	1 mM NaBu + 40 μ M Rev	5 mM NaBu + 40 μ M Rev	No cells - media only
B												
C												
D												
E												
F												
G												
H												

Figure 2.2

Seed one 96 well plate with 10.000 DLD-1 cells pr well. Volume of 200 μ L pr well.

Mastermix:
10ml with 500.000 cells

$$\frac{500.000}{1.131.180} = 0,442 \text{ cell sop} + 9,958 \text{ media}$$

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23/6-2026, Green team, Exp 2B.

200 μ l / well.

96-well plate.

~~Costa~~ and DLD1.

MC104
~~DMEM~~

200 μ l media

10 μ l 100 mM ResV
90 μ l DMSO $\rightarrow$ 100 μ l
10 mM ResV

MC104

~~DMEM~~
+
PBS + DMSO

1990 media

2 μ l PBS

8 μ l DMSO

1 mM Nabu
+
DMSO

1990 media

2 μ l Nabu

8 μ l DMSO

40 μ M ResV
+
PBS

1990 media

2 μ l PBS

8 μ l 40 μ M ResV 10 mM

40 μ M ResV
+
1 mM Nabu

1990 μ l Media

2 μ l Nabu

8 μ l ResV 10 mM

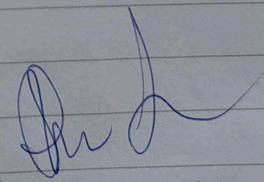
20 μ M ResV
+
1 mM Nabu

1990 μ l Media

4 μ l DMSO

2 μ l Nabu

4 μ l ResV. 10 mM.



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22.06.2026 green team: Line, Lars, Andreas
 CaCo-2
 Experiment 2b:
 Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
B	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
C	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
D	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
E	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
F	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
G	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
H	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
I	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
J	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
K	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄
L	Control	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄	10 mM NaH ₂ PO ₄	1 mM NaH ₂ PO ₄	2 mM NaH ₂ PO ₄	5 mM NaH ₂ PO ₄

Seed one 96 well plate with 10.000 CaCo-2 cells pr well. Volume of 200 uL

Mastermix:
 10ml with 500.000 cells

$$\frac{500.000 \text{ cells}}{195.075 \text{ cells/mL}} = 2.563 \text{ mL}$$

$$10 - 2.563 \text{ mL} = 7.436 \text{ mL media}$$

7-11 filled with CaCo2 cells (500.000 cells)

22/06/2026

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23/6-2026, Green Team, Exp 2B.

200 μ l / well.

96-well plate.

CACO2

DMEM

200 μ l media

10 μ l 100mM ResV
90 μ l DMSO

$\rightarrow$ 100 μ l
10mM
ResV

DMEM
+
PBS + DMSO

1990 media

2 μ l PBS

8 μ l DMSO

1mM Nabu
+
DMSO

1990 media

2 μ l Nabu

8 μ l DMSO

40 μ M ResV
+
PBS

1990 media.

2 μ l PBS

8 μ l 40 μ M ResV 10mM

40 μ M ResV
+
1mM Nabu

1990 μ l Media

2 μ l Nabu

8 μ l ResV

10mM

20 μ M ResV
1mM Nabu

1990 μ l Media

4 μ l DMSO

2 μ l Nabu.

4 μ l ResV.

10mM.

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23/6-2026, Green Team, Exp 2B. Caco2

200 μ l / well. 96-well plate.

DMEM  200 μ l media

DMEM + PBS + DMSO  1990 media
2 μ l PBS
8 μ l DMSO

1mM Nabu + DMSO  1990 media
2 μ l Nabu
8 μ l DMSO

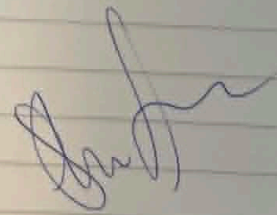
40 μ M ResV + PBS  1990 media
2 μ l PBS
8 μ l 40 μ M ResV 10 mM

40 μ M ResV + 1mM Nabu  1990 μ l Media
2 μ l Nabu
8 μ l ResV 10 mM

20 μ M ResV + 1mM Nabu  1990 μ l Media
4 μ l DMSO
2 μ l Nabu
4 μ l ResV 10 mM

10 μ l 100mM ResV $\rightarrow$ 100 μ l
90 μ l DMSO $\rightarrow$ 10 mM ResV





25.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b: As described for week 1
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. - incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

- a. Synergy HT, BIOTEK plate reader (28A.2-21)
- b. Excitation 530/25 nm (Filter wheel D) and
- c. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

- d. Synergy HT BioTek plate reader (28A.2-21)
- e. Excitation 360/40 nm (Filter wheel D)
- f. Emission 460/40 nm (Filter wheel A), Gain: 60.

~~cell check~~
~~Ctrl 1:~~

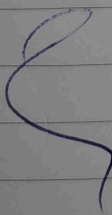
~~Ctrl 2:~~

~~CRISPR 1:~~

~~CRISPR 2:~~

~~MUICK 1:~~

~~MUICK 2:~~



25.06.2026 green team: Line, Lars, Andreas
CaCo-2:
Experiment 2b: As described for week 1
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
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